

# Continuous optimization

## ENT305A

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## 1 Reminders

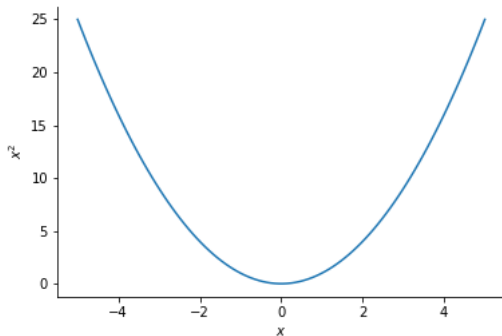
- Optimization problem
- Existence of a solution
- Gradient methods
- Newton's method

## 2 Optimality conditions for constrained problems

- Linear constraints
- Non-linear constraints
- Sensitivity analysis

$$f : x \rightarrow x^2, x \in [-5, 5]$$

$$\mathcal{D} = \mathbb{R}$$



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Let  $\bar{B}(\bar{x}, \varepsilon)$  denote the closed ball of center  $\bar{x}$  and radius  $\varepsilon$ .

A feasible point  $\bar{x}$  is a local solution to  $(P)$  if and only if there exists  $\varepsilon > 0$  such that  $\bar{x}$  is a **global** solution to the following **localized** problem:

$$\inf_{x \in \mathbb{R}^n} f(x), \quad x \in K \cap \bar{B}(\bar{x}, \varepsilon).$$

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Most of the time, the feasible set  $K$  is described by

$$K = \left\{ x \in \mathbb{R}^n \mid \begin{array}{l} h_i(x) = 0, \quad \forall i \in \mathcal{E} \\ g_j(x) \leq 0, \quad \forall j \in \mathcal{I} \end{array} \right\},$$

where  $h: \mathbb{R}^n \rightarrow \mathbb{R}^{m_1}$ ,  $g: \mathbb{R}^n \rightarrow \mathbb{R}^{m_2}$ .

We call the expressions

- $h_i(x) = 0$ : equality constraint
- $g_j(x) \leq 0$ : inequality constraint.

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- **Existence of a solution**
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## Existence of a solution

### Theorem 1 (existence of extreme value (Weierstrass))

Assume the following:

- ?
- ?

Then the optimization problem (P) has (at least) one solution.

## Existence of a solution

### Theorem 2 (existence of extreme value (Weierstrass))

Assume the following:

- $K$  is **non-empty** and **compact** (i.e. closed and bounded)
- $f$  is **continuous** on  $K$ .

Then the optimization problem (P) has (at least) one solution.

*Remarks.* If  $K = \{x \in \mathbb{R}^n \mid h_i(x) = 0, \forall i \in \mathcal{E}, g_j(x) \leq 0, \forall j \in \mathcal{I}\}$ , where  $h_i, g_j$  are continuous, then  $K$  is closed. In practical exercises, it is not necessary to justify the continuity of  $h_i$  or  $g_j$ .

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$$L = \mathbb{R} / \mathbb{Z} = \mathbb{R} / \mathbb{Z} \cdot 1 \quad \text{with} \quad \phi(D) = T \quad \text{and} \quad \phi(1) = 0 = \mathbb{Z} \cdot 1$$

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1. *Journal of the American Medical Association*, 1997; 277: 1001-1005.

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$$f(x) = \frac{1}{x^2} \quad f'(x) = -\frac{2}{x^3} \quad f''(x) = \frac{6}{x^4}$$

$$\frac{1}{2} \left( \frac{1}{2} \right) = \frac{1}{4}$$

1000

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- **Gradient methods**
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$$\inf_{x \in \mathbb{R}^n} f(x). \quad (P)$$
$$f(x_{k+1}) \leq f(x_k), \quad \forall k \in \mathbb{N},$$

How to compute  $x_{k+1}$ ?

A 3D plot of a function  $z = f(x, y)$  showing a saddle point at the origin  $(0, 0, 0)$ . The surface is colored with a gradient from blue (low values) to red (high values). The axes are labeled  $x$ ,  $y$ , and  $z$ . The  $z$ -axis ranges from 0 to 6, the  $x$ -axis from -2 to 2, and the  $y$ -axis from -2 to 2. The surface has a local minimum at the origin and local maxima along the  $x$  and  $y$  axes.

$$\inf_{x \in \mathbb{R}^n} f(x). \quad (P)$$
$$f(x_{k+1}) \leq f(x_k), \quad \forall k \in \mathbb{N},$$

How to compute  $x_{k+1}$ ?

---

Let  $x_k \in \mathbb{R}^n$ . Let  $d_k$  be a descent direction at  $x_k$ . Let  $\alpha > 0$ . Then

$$f(x_k + \alpha d_k) = f(x_k) + \underbrace{\alpha \langle \nabla f(x_k), d_k \rangle}_{\leq 0} + o(\alpha).$$

Therefore, if  $\alpha$  is small enough,

$$f(x_k + \alpha d_k) < f(x_k).$$

We can set

$$x_{k+1} = x_k + \alpha d_k.$$

THE

**P** **L** **I** **S** **T** **E**

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[illegible]

*On the choice of  $\alpha_k$ .*

Let  $x_k \in \mathbb{R}^n$  and let  $d_k$  be a descent direction. Define

$$\phi_k : \alpha \in \mathbb{R} \mapsto f(x_k + \alpha d_k).$$

Then

$$f(x_k + \alpha_k d_k) < f(x_k) \iff \phi_k(\alpha_k) < \phi_k(0).$$

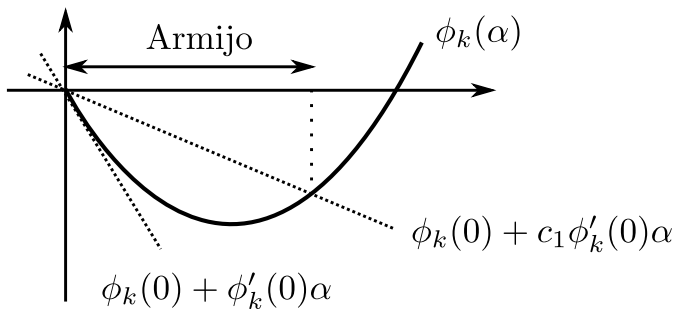
A natural choice would be an **exact line search**:

$$\alpha_k \in \arg \min_{\alpha \geq 0} \phi_k(\alpha).$$

and the quality of the step size.

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# Line-search methods

*Backtracking line search for the Armijo condition.*

- 1 Choose  $c_1 \in (0, 1)$ , an initial step  $\alpha_0 > 0$ , and a reduction factor  $\rho \in (0, 1)$ .

- 2 Set  $\alpha = \alpha_0$ .

- 3 While

$$\phi_k(\alpha) > \phi_k(0) + c_1 \alpha \phi'_k(0),$$

set

$$\alpha \leftarrow \rho \alpha.$$

- 4 Set  $\alpha_k = \alpha$ .

# Line-search methods

## Definition 6

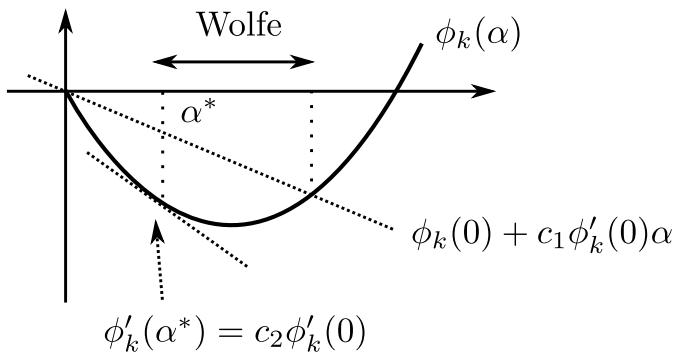
Let  $0 < c_1 < c_2 < 1$ . A step size  $\alpha > 0$  satisfies the **Wolfe conditions** if

$$\phi_k(\alpha) \leq \phi_k(0) + c_1 \alpha \phi'_k(0),$$

$$\phi'_k(\alpha) \geq c_2 \phi'_k(0).$$

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# Line-search methods

## *General comments.*

- If  $d_k$  is a descent direction and  $f$  is sufficiently smooth, an Armijo/Wolfe line search terminates after finitely many reductions.
- Without convexity, convergence to a global minimizer cannot in general be guaranteed. Under suitable assumptions, one typically obtains

$$\|\nabla f(x_k)\| \longrightarrow 0.$$

- Gradient methods may exhibit slow convergence, in particular for ill-conditioned problems.

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# Newton's method

*Main idea.*

Newton's method was originally introduced to solve nonlinear systems

$$F(x) = 0,$$

where  $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$  is continuously differentiable.

Given  $x_k$ , we use the linear approximation

$$F(x_k + d_k) \approx F(x_k) + DF(x_k)d_k.$$

The Newton step  $d_k$  is defined as a solution of

$$DF(x_k)d_k = -F(x_k).$$

If  $DF(x_k)$  is invertible, this gives

$$d_k = -DF(x_k)^{-1}F(x_k),$$

and the Newton iteration is

$$x_{k+1} = x_k + d_k.$$

# Newton's method

## *Remarks.*

- Suppose that  $F(\bar{x}) = 0$  and that  $DF(\bar{x})$  is invertible. Under the usual regularity assumptions, if  $x_0$  is sufficiently close to  $\bar{x}$ , Newton's iterates are well defined and converge **quadratically** to  $\bar{x}$ .
- If  $x_0$  is far from  $\bar{x}$ , convergence is not guaranteed.

## Newton's method

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Back to problem (P). Assume that  $f$  is continuously twice differentiable. Apply Newton's method with  $F(x) = \nabla f(x)$  so as to solve  $\nabla f(x) = 0$ . Update formula:

$$x_{k+1} = x_k - D^2f(x_k)^{-1}\nabla f(x_k).$$

The difficulties mentioned above are still relevant.

# Newton's method

We use the same general iteration as before:

$$x_{k+1} = x_k + \alpha_k d_k.$$

For the pure Newton method,

$$\alpha_k = 1,$$

and the Newton direction  $d_k$  solves

$$D^2f(x_k)d_k = -\nabla f(x_k).$$

If  $D^2f(x_k)$  is positive definite and  $\nabla f(x_k) \neq 0$ , then the Newton direction is a descent direction.

Indeed,

$$\begin{aligned}\langle \nabla f(x_k), d_k \rangle &= -\langle \nabla f(x_k), D^2f(x_k)^{-1} \nabla f(x_k) \rangle \\ &< 0.\end{aligned}$$

# Newton's method

- 1 Choose  $x_0 \in \mathbb{R}^n$ ,  $\varepsilon > 0$ , and a line-search strategy.
- 2 Set  $k = 0$ .
- 3 While

$$\|\nabla f(x_k)\| > \varepsilon,$$

do:

- (a) Compute a candidate Newton direction  $d_k$  by solving

$$D^2 f(x_k) d_k = -\nabla f(x_k).$$

- (b) If  $d_k$  is well defined and satisfies  $\langle \nabla f(x_k), d_k^N \rangle < 0$ , continue.  
Otherwise, use for instance the gradient direction

$$d_k = -\nabla f(x_k).$$

- (c) Choose  $\alpha_k > 0$  by a line search (Armijo, Wolfe), with  $\alpha_k = 1$  first.
- (d) Set

$$x_{k+1} = x_k + \alpha_k d_k.$$

- (e) Set  $k \leftarrow k + 1$ .

- 4 Output  $x_k$ .

# Newton's method

## *Comments.*

- Globalization strategies such as line search or trust-region methods improve the behavior of Newton's method far from a solution.
- Under suitable assumptions, a globalized Newton method generates iterates whose accumulation points are stationary.
- Near a nondegenerate local minimizer  $\bar{x}$ , local convergence can be recovered.
- Computing and storing  $D^2f(x_k)$  can be expensive. A dense Hessian requires  $O(n^2)$  storage, and solving a dense Newton system typically requires  $O(n^3)$  operations.

# Quasi-Newton methods

*Main idea.*

Quasi-Newton methods avoid computing the exact Hessian.

One may construct matrices  $H_k$  approximating the inverse Hessian:

$$H_k \approx D^2 f(x_k)^{-1}.$$

The search direction is then

$$d_k = -H_k \nabla f(x_k).$$

If  $H_k$  is positive definite and  $\nabla f(x_k) \neq 0$ , then

$$\langle \nabla f(x_k), d_k \rangle = -\langle \nabla f(x_k), H_k \nabla f(x_k) \rangle < 0,$$

so  $d_k$  is a descent direction.

Limited-memory variants such as L-BFGS can be implemented using  $O(n)$  storage when the memory parameter is fixed.

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# Constrained optimization problem

- We investigate in this section the problem

$$\inf_{x \in \mathbb{R}^n} f(x), \quad \text{s.t.} \quad \begin{cases} h_i(x) = 0, & \forall i \in \mathcal{E} \\ g_j(x) \leq 0, & \forall j \in \mathcal{I}. \end{cases} \quad (P)$$

- Let  $x \in \mathbb{R}^n$  be feasible. Let  $j \in \mathcal{I}$ . We say that
  - the inequality constraint  $j$  is **active** if  $g_j(x) = 0$
  - the inequality constraint  $j$  is **inactive** if  $g_j(x) < 0$ .
- *Remark.* All results of the section are true if  $\mathcal{E} = \emptyset$  or  $\mathcal{I} = \emptyset$ .

# Constrained optimization problem

Let us focus on  $g$ :

$$\inf_{x \in \mathbb{R}^n} f(x), \quad \text{s.t. } g(x) \leq 0. \quad (P)$$

**How to solve this as an unconstrained problem?**

We could instead of solving this problem (P) solve with  $\lambda \geq 0$

$$\begin{aligned} L(x, \lambda) &= f(x) + \langle \lambda, g(x) \rangle \\ &= f(x) + \lambda g(x). \end{aligned}$$

# Linear constraints

- Let  $f: \mathbb{R}^n \rightarrow \mathbb{R}$  and let  $h: \mathbb{R}^n \rightarrow \mathbb{R}^{m_1}$  and  $g: \mathbb{R}^n \rightarrow \mathbb{R}^{m_2}$  be  $C^2$ .
- Let the **Lagrangian**  $L: \mathbb{R}^n \times \mathbb{R}^{m_1} \times \mathbb{R}^{m_2} \rightarrow \mathbb{R}$  be defined by

$$\begin{aligned} L(x, \mu, \lambda) &= f(x) + \langle \mu, h(x) \rangle + \langle \lambda, g(x) \rangle \\ &= f(x) + \sum_{i=1}^{m_1} \mu_i h_i(x) + \sum_{j=1}^{m_2} \lambda_j g_j(x). \end{aligned}$$

The variables  $\mu, \lambda$  are referred to as **dual variables**.

# KKT conditions

Let  $\bar{x}$  be feasible for  $(P)$ .

The **Karush-Kuhn-Tucker (KKT) conditions** at  $\bar{x}$  are the existence of

$$(\mu, \lambda) \in \mathbb{R}^{m_1} \times \mathbb{R}^{m_2}$$

such that:

1 Stationarity:

$$\nabla_x L(\bar{x}, \mu, \lambda) = 0.$$

2 Dual feasibility:

$$\lambda_j \geq 0, \quad \forall j \in \mathcal{I}.$$

3 Complementarity:

$$\lambda_j g_j(\bar{x}) = 0, \quad \forall j \in \mathcal{I}.$$

4 Primal feasibility:

$$h_i(\bar{x}) = 0, \quad \forall i, \quad g_j(\bar{x}) \leq 0, \quad \forall j.$$

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# KKT conditions for affine constraints

## Theorem 7

*Assume that the constraint functions  $h$  and  $g$  are affine.*

*If  $\bar{x}$  is a **local solution** to  $(P)$ , then there exist*

$$(\mu, \lambda) \in \mathbb{R}^{m_1} \times \mathbb{R}^{m_2}$$

*such that  $\bar{x}$  satisfies the KKT conditions.*

Thus, for affine constraints,

$$\bar{x} \text{ local solution} \implies \bar{x} \text{ satisfies KKT.}$$

In general, the converse is **false**!

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# Exercise

Consider the problem

$$\min_{x \in \mathbb{R}} f(x) = -x^2$$

subject to the affine constraint

$$g(x) = x - 1 \leq 0.$$

- 1 Write the Lagrangian and the KKT conditions.
- 2 Find all pairs  $(x, \lambda)$  satisfying the KKT conditions.
- 3 For each KKT point  $x$ , determine whether it is a local solution of the constrained optimization problem.
- 4 Does the problem admit a global solution?

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# Exercise

We study the feasible point  $\bar{x} = 0$ .

The Lagrangian is

$$L(x, \lambda) = -x^2 + \lambda(x - 1).$$

The KKT conditions at  $\bar{x} = 0$  are:

■ **Primal feasibility:**

$$g(0) = -1 \leq 0.$$

■ **Stationarity:**

$$\nabla_x L(0, \lambda) = -2 \cdot 0 + \lambda = \lambda = 0.$$

Hence

$$\lambda = 0.$$

■ **Dual feasibility:**

$$\lambda = 0 \geq 0.$$

■ **Complementarity:**

$$\lambda g(0) = 0 \times (-1) = 0.$$

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## Exercise – Correction

For every  $x \neq 0$  sufficiently close to 0, we still have

$$x \leq 1,$$

so  $x$  is feasible.

$$f(x) = -x^2 < 0 = f(0).$$

Hence

$\bar{x} = 0$  is not a local minimizer,

but a **local maximizer** on the feasible set.

# Linear constraints

**Example 1.** Case of **one equality constraint**:

$$m = 1, \quad \mathcal{E} = \{1\}, \quad \mathcal{I} = \emptyset.$$

Let  $h(x) = Ax - b$ . The matrix  $A$  is a row vector, and  $q = A^\top = \nabla h(x)$ .

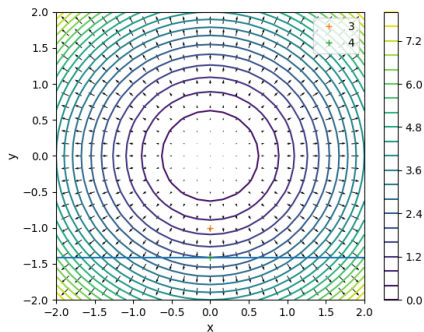
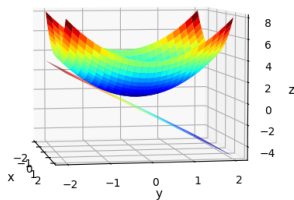
*Proof of KKT conditions.*

- We have:

$$\nabla_x L(\bar{x}, \mu) = \nabla f(\bar{x}) + \mu \nabla h(\bar{x}) = \nabla f(\bar{x}) + \mu q = 0.$$

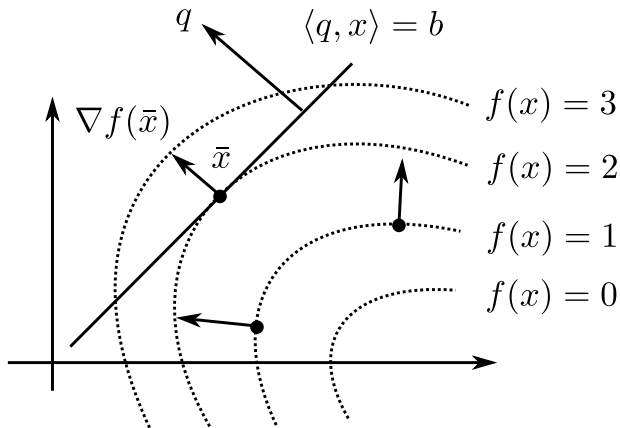
- Let  $\mu \in \mathbb{R}$  be such that  $\nabla f(\bar{x}) = -\mu q$ .
- Geometrically,  $\nabla f(\bar{x})$  and  $q$  are **collinear**.

# KKT stationarity



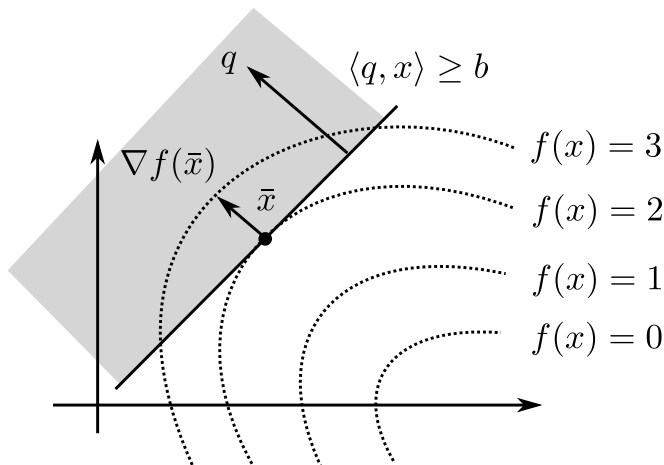
# Linear constraints

$$q = \nabla h$$



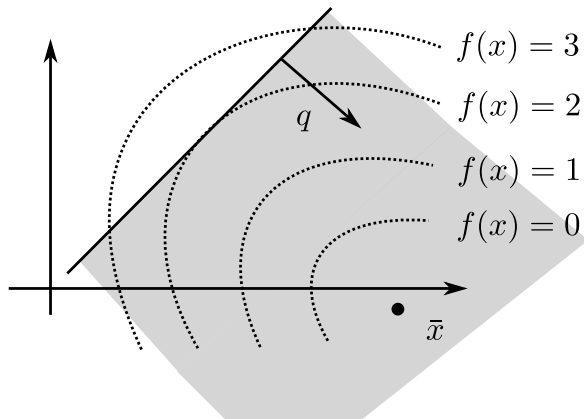
# Linear constraints

$q = \nabla g$  **Example 2(a).** Case of **one active inequality constraint**:



# Linear constraints

$q = \nabla g$  **Example 2(b).** Case of **one inactive inequality constraint**:



# KKT conditions for affine constraints

## Remarks.

- A pair  $(\mu, \lambda)$  satisfying the KKT conditions is called a vector of **Lagrange multipliers** associated with  $\bar{x}$ .
- Further assumptions are required to have uniqueness of  $(\mu, \lambda)$ .
- If  $\mathcal{I} = \emptyset$ , then the sign condition and the complementarity conditions are trivially satisfied.
- The theorem also applies when the number of constraints exceeds the number of variables.

$\lambda, \mu =$  Penalization !!

# Lagrange multiplier as a marginal value

Consider

$$V(\textcolor{red}{b}) = \min_x f(x) \quad \text{s.t.} \quad a(x) \leq b,$$

with Lagrangian

$$L(x, \lambda; b) = f(x) + \lambda(a(x) - b), \quad \lambda \geq 0.$$

Assume that the constraint is active and that the solution  $x^*(b)$  varies smoothly with  $b$ . The KKT conditions give

$$V'(b) = -\lambda.$$

# Lagrange multiplier as a marginal value

**Proof.**

$$\nabla f(x^*) + \lambda \nabla a(x^*) = 0, \quad a(x^*) = b.$$

Since  $V(b) = f(x^*(b))$ ,

$$V'(b) = \nabla f(x^*)^\top x^{*'}(b).$$

Using the stationarity condition,

$$V'(b) = -\lambda \underbrace{\nabla a(x^*)^\top x^{*'}(b)}_{=1}$$

Thus,  $V'(b) = -\lambda.$

Hence, for a small increase  $\delta b$ ,

$$V(b + \delta b) \simeq V(b) - \lambda \delta b.$$

# Energy example: the marginal value of capacity

A demand  $d$  must be supplied. Local energy costs 10 €/MWh and is limited by  $p \leq b$ . The remaining energy is bought from the grid at 30 €/MWh:

$$V(b) = \min_{\substack{0 \leq p \leq d \\ p \leq b}} [10p + 30(d - p)].$$

Therefore,

$$p^* = \min(d, b)$$

and

$$V(b) = \begin{cases} 30d - 20b, & b < d, \\ 10d, & b \geq d. \end{cases}$$

# Energy example: the marginal value of capacity

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$$V(b) = \begin{cases} 30d - 20b, & b < d, \\ 10d, & b \geq d. \end{cases}$$

**Capacity active:**  $b < d$

$$p^* = b, \quad \lambda = 20.$$

As long as  $b + \delta b \leq d$ ,

$$V(b + \delta b) - V(b) = -20 \delta b.$$

**Capacity inactive:**  $b > d$

$$p^* = d < b, \quad \lambda = 0.$$

All the demand is already supplied by local energy.

# Lagrangian formulation

*Main idea of the Lagrangian formulation:*

Consider the primal problem (P):  $\inf_{x \in \mathbb{R}^n} f(x)$  s.t.:  $g(x) \leq 0$

(with  $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$ ).

Let  $p^*$  be the optimal cost of (P), and  $L$  the associated Lagrangian formulation. Then  $p^* = \inf_{x \in \mathbb{R}^n} \sup_{\lambda \in \mathbb{R}_+^m} L(x, \lambda)$ . Indeed,

$$\begin{aligned} \sup_{\lambda \in \mathbb{R}_+^m} L(x, \lambda) &= \sup_{\lambda \in \mathbb{R}_+^m} \left( f(x) + \sum_{j=1}^m \lambda_j g_j(x) \right), \\ &= \begin{cases} f(x) & \text{if } g_j(x) \leq 0, \forall j = 1, \dots, m \\ \infty & \text{otherwise (if } x \text{ not feasible and } g_j(x) > 0), \end{cases} \end{aligned}$$

# Lagrangian formulation

*Main idea of the Lagrangian formulation:*

And we always have  $L(x, \lambda) \leq f(x)$  for an admissible point  $x$ . So

$$L(x^*, \lambda) \leq \inf_x \sup_{\lambda} L(x, \lambda)$$

and

$$d^* = \sup_{\lambda} \inf_x L(x, \lambda) \leq \inf_x \sup_{\lambda} L(x, \lambda) = p^*$$

We have strong duality if

$$d^* = p^*$$

# Linear constraints

## Exercise.

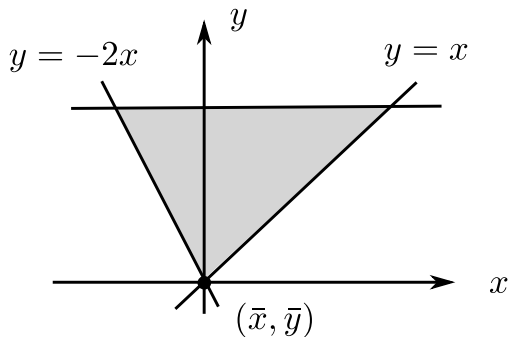
Consider the problem

$$\inf_{(x,y) \in \mathbb{R}^2} f(x,y) := y, \quad \text{s.t.} \quad \begin{cases} g_1(x,y) := -2x - y \leq 0, \\ g_2(x,y) := x - y \leq 0, \\ g_3(x,y) := y - 3 \leq 0. \end{cases}$$

- Draw the feasible set and find (geometrically) the solution.
- Verify that the KKT conditions are satisfied.

# Linear constraints

*Solution.*



# Linear constraints

- Solution to the problem:  $(\bar{x}, \bar{y}) = (0, 0)$ .
- Let  $\lambda \in \mathbb{R}^3$  be the associated Lagrange multiplier.  
Necessarily  $\lambda_3 = 0$ , since  $g_3(\bar{x}, \bar{y}) < 0$ , by complementarity.
- Lagrangian:

$$L(x, y, \lambda) = y - \lambda_1(2x + y) - \lambda_2(-x + y).$$

- The stationarity condition yields:

$$0 = \frac{\partial L}{\partial x}(0, 0) = -2\lambda_1 + \lambda_2$$

$$0 = \frac{\partial L}{\partial y}(0, 0) = 1 - \lambda_1 - \lambda_2.$$

# Linear constraints

- This linear system has a unique solution

$$\lambda_1 = 1/3 \geq 0 \quad \lambda_2 = 2/3 \geq 0.$$

The sign condition is satisfied.

## 1 Reminders

- Optimization problem
- Existence of a solution
- Gradient methods
- Newton's method

## 2 Optimality conditions for constrained problems

- Linear constraints
- Non-linear constraints
- Sensitivity analysis

# Non-linear constraints

## Definition 8

Let  $\bar{x}$  be a feasible point. Let the set of **active inequality constraints**  $\mathcal{I}_0(\bar{x})$  be defined by

$$\mathcal{I}_0(\bar{x}) = \{j \in \mathcal{I} \mid g_j(\bar{x}) = 0\}.$$

We say that the **Linear Independence Constraint Qualification (LICQ)** holds at  $\bar{x}$  if the following set of vectors is linearly independent:

$$\{\nabla h_i(\bar{x})\}_{i \in \mathcal{E}} \cup \{\nabla g_j(\bar{x})\}_{j \in \mathcal{I}_0(\bar{x})}$$

# Non-linear constraints



# Non-linear constraints

## Theorem 9

Let  $\bar{x}$  be a **local solution** to  $(P)$ . Assume that the LICQ holds at  $\bar{x}$ . Then there exists a unique  $(\mu, \lambda)$  such that the **KKT conditions** are satisfied.

*Remarks.*

- When LICQ fails, numerical methods may have difficulty identifying stable multipliers.

# Non-linear constraints

## Example 4.

Consider the problem

$$\inf_{x \in \mathbb{R}} x, \quad \text{subject to: } x^2 \leq 0.$$

The unique feasible point is  $\bar{x} = 0$ , hence it is the solution.

Lagrangian:

$$L(x, \lambda) = x + \lambda x^2.$$

At zero:

$$\nabla_x L(0, \lambda) = 1 + 2\lambda \bar{x} = 1 \neq 0.$$

The LICQ is not satisfied, since  $\nabla g_1(0) = 0$ .

# Non-linear constraints

## Theorem 10

*Assume that*

- $f$  is **convex**
- for all  $i \in \mathcal{E}$ , the map  $x \mapsto h_i(x)$  is **affine**
- for all  $j \in \mathcal{I}$ , the map  $x \mapsto g_j(x)$  is **convex**.

*Then any feasible point  $\bar{x}$  satisfying the **KKT conditions** is a **global solution** to the problem.*

*Remark.* The result holds whether the LICQ holds or not at  $\bar{x}$ .

# Exercise

**Exercise.** Consider the function  $f: (x, y) \in \mathbb{R}^2 \mapsto \exp(x + y^2) + y + x^2$ .

- 1 Prove that  $f$  is coercive. *Indication:* Use  $\exp(z) \geq 1 + z$
- 2 Compute  $\nabla f(x, y)$  and  $\nabla^2 f(x, y)$ .
- 3 We recall that a symmetric matrix of size 2 of the form  $\begin{pmatrix} a & b \\ b & c \end{pmatrix}$  is positive semidefinite if and only if  $a + c \geq 0$  and  $ac - b^2 \geq 0$ . Using this fact, prove that  $f$  is convex.
- 4 We consider the following problem:

$$\inf_{(x,y) \in \mathbb{R}^2} f(x, y), \quad \text{subject to: } \begin{cases} -x - y \leq 0 \\ -x - 2 \leq 0. \end{cases} \quad (\mathcal{P})$$

Verify that  $(0, 0)$  is feasible and satisfies the KKT conditions.

- 5 Is the point  $(0, 0)$  a global solution to problem  $(\mathcal{P})$ ?

# Exercise

*Solution.*

1. We use the inequality:  $\exp(z) \geq 1 + z$ , which yields:

$$\begin{aligned}
 f(x, y) &\geq x + y^2 + y + x^2 \\
 &= \frac{1}{2}(x^2 + y^2) + \frac{1}{2}(x^2 + 2x + 1) + \frac{1}{2}(y^2 + 2y + 1) - 1 \\
 &= \frac{1}{2}\|(x, y)\|^2 + \frac{1}{2}(x + 1)^2 + \frac{1}{2}(y + 1)^2 - 1 \xrightarrow{\|(x, y)\| \rightarrow \infty} \infty.
 \end{aligned}$$



# Exercise

3. Proof that  $D^2f$  is positive semidefinite. It holds:

$$a + c = (3 + 4y^2) \exp(x + y^2) + 2 \geq 0$$

and

$$ac - b^2 = 2 \exp(2x + 2y^2) + 4(1 + 2y^2) \exp(x + y^2) \geq 0.$$

It follows that  $D^2f(x, y)$  is positive semidefinite, for all  $(x, y)$ .  
Therefore  $f$  is a convex function.

## Exercise

- KKT conditions. Lagrangian:

$$L(x, y, \lambda_1, \lambda_2) = \exp(x + y^2) + y + x^2 - \lambda_1(x + y) - \lambda_2(x + 2).$$

Therefore,

$$\frac{\partial L}{\partial x}(0, 0, \lambda_1, \lambda_2) = 1 - \lambda_1 - \lambda_2, \quad \frac{\partial L}{\partial y}(0, 0) = 1 - \lambda_1.$$

Taking  $\lambda_1 = 1$  and  $\lambda_2 = 0$ , we have:

- 1 Stationarity:  $\frac{\partial L}{\partial x}(0, 0, 1, 0) = \frac{\partial L}{\partial y}(0, 0, 1, 0) = 0$ .
- 2 Sign condition:  $\lambda_1 \geq 0, \lambda_2 \geq 0$ .
- 3 Complementary: the second constraint is inactive and the corresponding Lagrange multiplier is null.

## Exercise

5. We have the following:
- The cost function is convex.
  - The functions  $-(x + y)$  and  $-(x + 2)$  are convex.
  - The point  $(0, 0)$  is feasible and satisfies the KKT conditions.

Therefore  $(0, 0)$  is a global solution.

# Course summary

|                                  | Necessary conditions | Sufficient conditions                                                         |
|----------------------------------|----------------------|-------------------------------------------------------------------------------|
| Abstract formulation<br>(exist.) |                      | if $K$ is non-empty and compact, $f \in C^0(K)$<br>then at least one solution |
|                                  |                      | if $K$ closed,<br>$f \in C^0(K)$ , coercive<br>then at least one solution     |

|                                             | Necessary conditions                                                                                 | Sufficient conditions                                                                                                  |
|---------------------------------------------|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| No constraints<br>$K = \mathbb{R}^d$ (opt.) | if $\bar{x}$ local sol.,<br>$f \in C^2(\mathbb{R}^d)$ then,<br>$D^2f(\bar{x})$ is positive semi-def. | if $f \in C^2(\mathbb{R}^d)$ , $\nabla f(\bar{x}) = 0$ ,<br>$D^2f(\bar{x})$ positive def.<br>then $\bar{x}$ local sol. |
| Affine constraints                          | $\bar{x}$ local sol. then KKT                                                                        | $f$ convex,<br>then KKT=global sol.                                                                                    |
| Non-linear constraints                      | $\bar{x}$ local sol., LICQ then KKT                                                                  | $f$ convex,<br>$h$ affine, $g$ convex,<br>then KKT=global sol.                                                         |

## **Temporary page!**

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